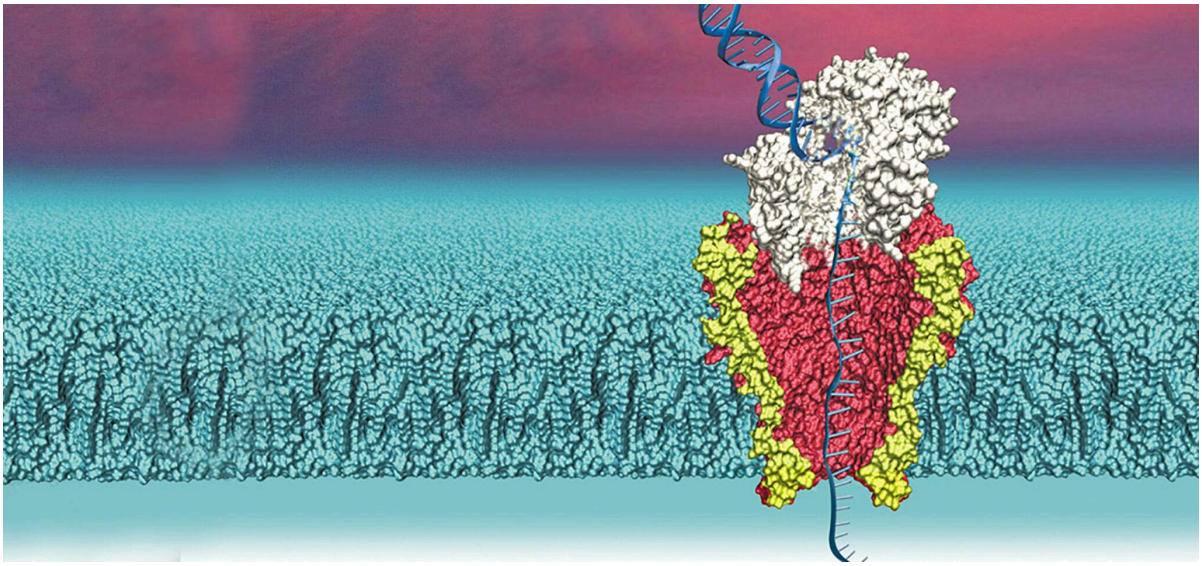
# Chapter 20

*Campbell Biology, AP Edition, 12e*

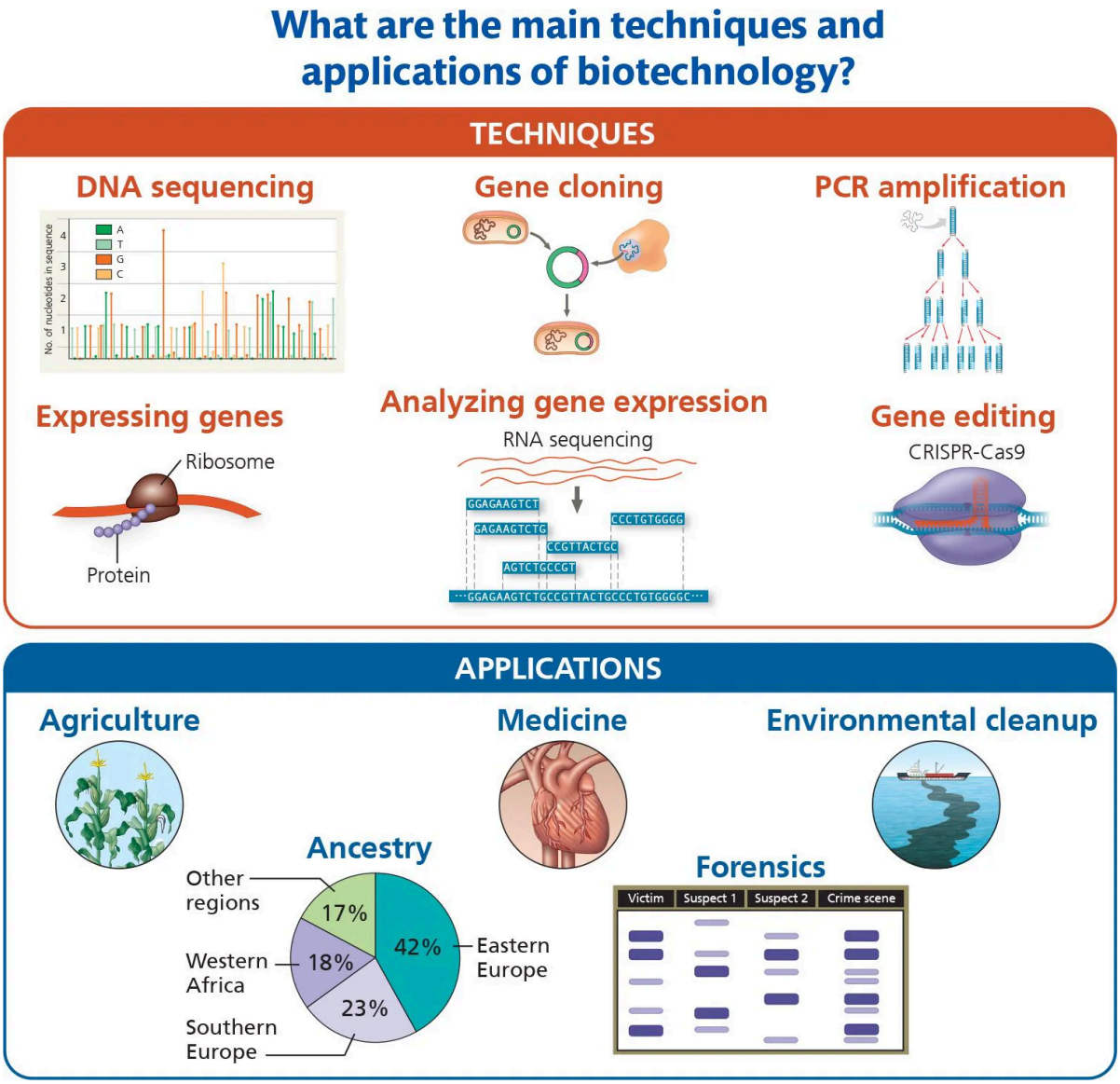


# Chapter 20

# DNA Tools and Biotechnology

**Figure 20.1**

This model shows a technique in which a DNA strand is passed through a small pore in a membrane. The resulting changes in an electrical current are used to determine the nucleotide sequence. The first human genome sequence, completed in 2003, took 13 years and cost $1 billion; the time and cost of genome sequencing have been greatly reduced by improved methods like the one shown here.




Techniques include the following. D N A sequencing, gene cloning, P C R amplification, expressing genes, analyzing gene expression, and gene editing. Applications include the following. Agriculture, medicine, environmental cleanup, ancestry, and forensics.

## Key Concepts

- **20.1DNA sequencing and DNA cloning are valuable tools for genetic engineering and biological inquiry**
- **20.2Biologists use DNA technology to study gene expression and function**
- **20.3Cloned organisms and stem cells are useful for basic research and other applications****

- **20.4The practical applications of DNA-based biotechnology affect our lives in many ways**

Biotechnology makes it possible to identify and manipulate DNA and RNA in order to make useful products **(BIG IDEA 3)**. Scientists can isolate, sequence, and transfer genes from one organism to another and thereby delineate the relationships of organisms, solve medical and criminal mysteries, and increase agricultural productivity **(BIG IDEA 4)**.

## Study Tip

**Apply what you've learned:** Write down a gene-related question you've wondered about. Then make a table listing techniques you could use to investigate your question and how you would apply them. Below is an example.

| Is autism caused by genes, the environment, or both? | |
|---|---|
| Technique | Application to question |
| DNA sequencing | -Compare DNA sequences from people with autism with sequences from nonaffected people. |

The table has columns titled, Technique, and Application to question. The technique is D N A sequencing. The application is as follows: Compare D N A sequences from people with autism with sequences from non-affected people.

## Get Ready for This Chapter

In this chapter, you will learn about DNA sequencing and related DNA techniques; the techniques used to study the expression and function of genes; the use of cloned organisms and stem cells; and the various applications of DNA-based technology. Before beginning this chapter, you should be able to do the following:

- Describe the process of gene expression (see Concept 17.1 and Figure 17.4).
- Distinguish between the processes of gene expression in a bacterial cell and a eukaryotic cell (see Concept 17.1 and Concept 17.4).
- Describe the structure of DNA (see Concept 5.5, Concept 16.1, and Figure 16.23).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following events occurs during the process of translation?

```
An RNA is synthesized.
```

That's incorrect. Try again.

**Incorrect.** Review the following material: Translation is the synthesis of a polypeptide using the information in mRNA. In semiconservative replication, a DNA strand serves as a template for making a new complementary DNA strand, and transcription is the synthesis of RNA using the information in DNA. Post-

translational modifications may be required to further modify proteins before the protein is fully functional in the cell.

```
A polypeptide is synthesized.
```

That's correct.

**Correct!** If you had trouble with this question, review the following material: Translation is the synthesis of a polypeptide using the information in mRNA. In semiconservative replication, a DNA strand serves as a template for making a new complementary DNA strand, and transcription is the synthesis of RNA using the information in DNA. Post-translational modifications may be required to further modify proteins before the protein is fully functional in the cell.

```
A protein is modified.
```

That's incorrect. Try again.

**Incorrect.** Review the following material: Translation is the synthesis of a polypeptide using the information in mRNA. In semiconservative replication, a DNA strand serves as a template for making a new complementary DNA strand, and transcription is the synthesis of RNA using the information in DNA. Post-translational modifications may be required to further modify proteins before the protein is fully functional in the cell.

```
DNA is replicated.
```

That's incorrect. Try again.

**Incorrect.** Review the following material: Translation is the synthesis of a polypeptide using the information in mRNA. In semiconservative replication, a DNA strand serves as a template for making a new complementary DNA strand, and transcription is the synthesis of RNA using the information in DNA. Post-translational modifications may be required to further modify proteins before the protein is fully functional in the cell.

2. If you were to observe gene expression in a cell, which of the following clues would indicate that the cell is prokaryotic?

```
There are mRNAs that are being translated at the same time as they are being t
```

That's correct.

**Correct!** If you had trouble with this question, review the following material: In both bacteria and eukaryotes, multiple ribosomes, called polyribosomes, translate a single mRNA at the same time. Additionally, both bacteria and eukaryotes augment the number of copies of a polypeptide by transcribing multiple mRNAs from the same gene. In the absence of a nucleus, however, prokaryotic cells can simultaneously transcribe and translate the same gene, whereas in eukaryotic cells, the nuclear envelope segregates transcription from translation. RNA processing is unique to eukaryotic cells, during which spliceosomes carry out RNA splicing.

```
A single mRNA has many ribosomes bound to it.
```

That's incorrect. Try again.

**Incorrect.** Review the following material: In both bacteria and eukaryotes, multiple ribosomes, called polyribosomes, translate a single mRNA at the same time. Additionally, both bacteria and eukaryotes augment the number of copies of a polypeptide by transcribing multiple mRNAs from the same gene. In the absence of a nucleus, however, prokaryotic cells can simultaneously transcribe and translate the same gene, whereas in eukaryotic cells, the nuclear envelope segregates transcription from translation. RNA processing is unique to eukaryotic cells, during which spliceosomes carry out RNA splicing.

```
A single gene has many RNA polymerases bound to it.
```

That's incorrect. Try again.

**Incorrect.** Review the following material: In both bacteria and eukaryotes, multiple ribosomes, called polyribosomes, translate a single mRNA at the same time. Additionally, both bacteria and eukaryotes augment the number of copies of a polypeptide by transcribing multiple mRNAs from the same gene. In the absence of a nucleus, however, prokaryotic cells can simultaneously transcribe and translate the same gene, whereas in eukaryotic cells, the nuclear envelope segregates transcription from translation. RNA processing is unique to eukaryotic cells, during which spliceosomes carry out RNA splicing.

```
Spliceosomes are carrying out RNA splicing.
```

That's incorrect. Try again.

**Incorrect.** Review the following material: In both bacteria and eukaryotes, multiple ribosomes, called polyribosomes, translate a single mRNA at the same time. Additionally, both bacteria and eukaryotes augment the number of copies of a polypeptide by transcribing multiple mRNAs from the same gene. In the

absence of a nucleus, however, prokaryotic cells can simultaneously transcribe and translate the same gene, whereas in eukaryotic cells, the nuclear envelope segregates transcription from translation. RNA processing is unique to eukaryotic cells, during which spliceosomes carry out RNA splicing.

3. Which of the following accurately describes the structure of DNA?

```
A DNA molecule is a polynucleotide that is primarily single-stranded, and suga
```

That's incorrect. Try again.

**Incorrect.** Review the following material: DNA molecules have two polynucleotides, or “strands,” that wind around an imaginary axis, forming a **double helix**. In each strand, adjacent nucleotides are joined by sugar-phosphate bonds, and the two strands are held together by hydrogen bonds between the paired bases on opposite strands. A polypeptide consists of a chain of amino acids held together by peptide bonds between adjacent amino acids.

```
A DNA molecule consists of two strands of polynucleotides that form a double helix;
```

That's incorrect. Try again.

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```
A DNA molecule consists of two strands of amino acids that form a double helix; adj
```

That's incorrect. Try again.

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```
A DNA molecule consists of two strands of polynucleotides that form a double helix;
```

That's correct.

**Correct!** If you had trouble with this question, review the following material: DNA molecules have two polynucleotides, or “strands,” that wind around an imaginary axis, forming a **double helix**. In each strand, adjacent nucleotides are joined by sugar-phosphate bonds, and the two strands are held together by hydrogen bonds between the paired bases on opposite strands. A polypeptide consists of a chain of amino acids held together by peptide bonds between adjacent amino acids.

## Learning Objectives

After studying this chapter, you should be able to do the following:

- 20.1Describe DNA sequencing, DNA cloning and the polymerase chain reaction.
- 20.2Identify techniques that allow us to study the expression and function of one or more genes.
- 20.3Explain how multicellular organisms can be cloned, and identify ways to obtain animal stem cells, then describe the uses of stem cells.
- 20.4List some practical applications of DNA-based technology.